

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**  
**Sixth Semester of B. Pharm. Examination**  
**University Theory Examination October 2015**  
**BIOPHARMACEUTICALS (PH312)**

**Date: 15/10/2015, Thursday      Time: 10:00 a.m. to 01:00 p.m.      Maximum Marks: 80**

**Instructions:**

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

**SECTION- I**

- Q 1** Attempt any **TWO** of the following.
- |          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>A</b> | 1. Write note on requirements of an ideal blood donor.                   | <b>05</b> |
|          | 2. Discuss the various steps involved in the production of streptomycin. | <b>05</b> |
| <b>B</b> | 1. Write note on human normal immunoglobulin.                            | <b>05</b> |
|          | 2. Discuss the various steps involved in the production of vitamin B12.  | <b>05</b> |
| <b>C</b> | 1. Write note on human thrombin.                                         | <b>05</b> |
|          | 2. Write note on sterilization of fermenter.                             | <b>05</b> |
- Q 2** Attempt any **TWO** of the following.
- |          |                                                                                   |           |
|----------|-----------------------------------------------------------------------------------|-----------|
| <b>A</b> | 1. Define fermentation. Discuss the control of different parameters of fermenter. | <b>05</b> |
|          | 2. Write note on human fibrinogen.                                                | <b>05</b> |
| <b>B</b> | 1. Write note on role of biotechnology for producing diagnostic aids.             | <b>05</b> |
|          | 2. Write note on screening of antibiotic producing microorganisms.                | <b>05</b> |
| <b>C</b> | 1. Write note on ELISA diagnostic kit.                                            | <b>05</b> |
|          | 2. Write note on dried human plasma.                                              | <b>05</b> |

**SECTION- II**

- Q 3** Attempt any **TWO** of the following.
- |          |                                                                                                                       |           |
|----------|-----------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | Explain Antigen-Antibody reaction in detail.                                                                          | <b>10</b> |
| <b>B</b> | Differentiate active & Passive Immunity. Write short note on preparation, standardization and storage of BCG Vaccine. | <b>10</b> |
| <b>C</b> | Classify Types of allergy. Explain diagnosis and treatment of allergy.                                                | <b>10</b> |
- Q 4** Attempt any **TWO** of the following.
- |          |                                                                                                      |           |
|----------|------------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | What is polyclonal & monoclonal antibodies? Describe hybridoma technology for monoclonal antibodies. | <b>10</b> |
| <b>B</b> | Differentiate absorbable and nonabsorbable sutures. Detail note on catgut.                           | <b>10</b> |
| <b>C</b> | Detail note on standardization, packaging and labeling of surgical products.                         | <b>10</b> |